

FC6P01 Project

Proposal

Sprint & Capacity Management Software

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Date: Sunday, 7th December 2025

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Project details

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Date Submitted : 7th December, 2025

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Introduction

Software development teams often operate within fast paced environments where multiple projects, clients, and deadlines compete for attention. As organizations grow, managing developer workload becomes increasingly difficult, especially when team members contribute to several projects at the same time. Without a structured method for task allocation and sprint planning, teams experience issues such as unclear workload distribution, unexpected delays, frequent context switching, and difficulty handling urgent tasks. These challenges create bottlenecks that directly impact productivity and overall project delivery.

The problem addressed in this project is the lack of a unified system that can track developer capacity, manage sprint commitments, and handle adhoc tasks in a predictable and organized manner. In many workplaces, developers are assigned tasks through verbal communication or multiple disconnected tools, which results in inaccurate workload visibility. When urgent tasks arise, managers struggle to understand how reallocations will affect current sprint goals, often leading to over allocation and missed deadlines.

This project proposes the development of a web based task and sprint management system that uses modern full stack technologies such as React, Node.js, and PostgreSQL to provide real time visibility into developer workload. The solution will offer features for capacity tracking, sprint planning, task assignment, and handling unplanned work, enabling teams to make data driven decisions and improve the efficiency of their development process.

Background and Motivation

Software development teams often deal with complex workloads, shifting priorities, and pressure to meet strict delivery timelines. In many organizations, tasks are allocated verbally or across multiple disconnected tools, which leads to issues such as duplicated work, unclear ownership, and unbalanced workloads. Research in modern Agile environments shows that ineffective task visibility directly increases context switching and reduces team productivity. This highlights the need for systems that can streamline sprint planning and capacity management in a structured and predictable way.

In my workplace, one of the most common challenges is the lack of a unified view of developer availability. When multiple clients and projects are handled at the same time, it becomes difficult to track who is fully allocated and who still has capacity. This becomes even more problematic when urgent or unplanned tasks appear during a sprint, because there is no reliable system to understand how reallocations will affect deadlines or work quality.

This problem domain is worth studying because it directly affects the efficiency, performance, and well-being of development teams. A workload imbalance has been shown to reduce software quality and increase burnout in long running teams. By exploring a better method of capacity tracking, sprint management, and task handling, teams can operate with fewer bottlenecks and make more informed decisions when assigning work.

I am motivated to carry out this project because I have personally experienced these issues while working with internal teams and clients. With my background in web development and full stack technologies, I have the capability to design and implement a practical system that can be used in a real organizational environment. This project also allows me to apply my knowledge of modern frameworks, database design, and user centered application architecture.

Problem in Brief

The primary problem addressed in this project is the absence of an effective and centralized method to manage developer capacity and task allocation in a multi project environment. When developers are working across several clients at once, it becomes difficult to maintain clarity on who is available, who is overloaded, and how sprint commitments are being tracked. This lack of visibility results in poor task distribution and delays in delivery.

A related issue is the handling of unplanned or adhoc tasks. These tasks commonly arise in software development and often disrupt ongoing sprint commitments. Without a structured way to evaluate team workload in real time, managers are forced to assign these tasks without understanding the actual impact on deadlines. This leads to over allocation and constant reprioritization.

Solving this problem is valuable because it improves project predictability, reduces wasted time, and helps teams make data driven decisions about workload and planning. A well designed system will allow teams to maintain clearer communication and reduce the risk of sprint failure.

Aim

The aim of this project is to develop a web based system that manages developer capacity and sprint tasks using modern full stack technologies.

Objectives

- To conduct a critical review of existing challenges in developer workload management, sprint planning, and adhoc task allocation within software development environments.
- To carry out a critical study of full stack technologies, frameworks, and architectural approaches that can support the development of a sprint and capacity management system.
- To design and develop a functional web based system that enables task allocation, capacity tracking, sprint management, and real time workload visibility.
- To evaluate the performance and usability of the proposed system through testing, feedback, and comparison with existing methods used in team environments.
- To prepare comprehensive final documentation that includes system design, implementation details, evaluation outcomes, and overall project reflections.

Proposed Solution

The proposed system is a web based task and sprint management application that provides a centralized platform for allocating developers to tasks, monitoring capacity, and planning sprints. The solution will introduce a structured method for visualizing workload distribution using capacity charts and sprint timelines.

The architecture will follow a modern full stack pattern. The frontend will be developed using a reactive framework such as React or Next.js to create an interactive user experience. The backend will be built with Node.js and a REST or GraphQL API to handle task creation, user management, sprint configuration, and capacity calculations. The system will store its data in a relational database such as PostgreSQL, allowing efficient modeling of tasks, users, sprints, and capacity metrics.

High level diagrams such as use case diagrams, data flow diagrams, and architecture diagrams will be included in the full documentation to illustrate the overall structure of the system. These diagrams will show how the frontend, backend, and database interact and how data flows between different components.

The system will also support features such as task status updates, progress tracking, adhoc task handling, and visual workload dashboards. Through these features, teams will be able to manage planned and unplanned work more effectively while maintaining transparency across the entire development process.

Resource Requirements

The development of the Sprint and Capacity Management Software will require both hardware and software resources. The project will be developed using a personal computer with sufficient processing power to run development tools, a modern web browser, and local server environments. A machine with at least an Intel i5 or equivalent processor, 8 GB of RAM, and stable internet connectivity will be adequate for development and testing activities.

The software requirements include a full stack development environment. The frontend will be developed using React or Next.js, while the backend will use Node.js along with a suitable API framework. PostgreSQL will be used as the primary database. Additional tools such as Visual Studio Code, Git for version control, and Postman for API testing will also be required. These tools are freely available and do not contribute additional cost to the project.

For deployment and demonstration purposes, a cloud hosting platform such as Render, Railway, or Vercel may be used. These platforms offer free or low cost tiers that are sufficient for academic projects. Diagramming tools such as Draw.io or Figma may be used to create system architecture diagrams and design illustrations. Overall, the estimated cost for this project remains minimal, as most of the required software and services are free or available under community plans.

Deliverables

The final deliverables of this project will consist of both the completed software system and all supporting documentation.

The primary deliverable is a fully functional web based Sprint and Capacity Management Software that includes features for task allocation, sprint planning, capacity tracking, and handling adhoc work. The system will include user interfaces for administrators and developers, along with the necessary backend services and database structures.

Comprehensive project documentation will also be provided. This will include requirement specifications, design diagrams, system architecture descriptions, implementation details, testing reports, and evaluation results. A user guide will be prepared to explain how to operate the system and its core features. Additional materials such as a presentation or demonstration video may be included to support the final evaluation and assist users in understanding the system workflow.

Suggested Starting Point

The first step in developing this project will be to conduct a detailed review of the problem domain, focusing on existing sprint management practices, developer capacity tracking methods, and the limitations of current tools used in team environments. This will help refine the functional requirements and ensure that the system addresses the specific challenges identified.

Next, the project will begin with requirement gathering and the creation of high level design documents. This includes drafting use case diagrams, defining core system modules, and designing the database schema. Establishing a solid architectural foundation at the beginning will make the development process more structured and manageable.

Following the design phase, the development environment will be set up using the selected technologies. Initial work will focus on backend API development and database configuration, followed by frontend interfaces for task management and capacity visualization. Once core features are implemented, testing and refinement will take place to ensure the system functions as expected.

Project Plan

A	B	C	D	E	F	G	H	I	J
Task	December, 2025	January, 2026	February, 2026	March, 2026	April, 2026	May, 2026	June, 2026	July, 2026	August, 2026
Review and Problem Analysis									
Requirement Gathering and Finalization									
High Level System Design									
Database Design and Architecture									
Backend Development (APIs, Auth, Task Management)									
Frontend Development (UI, Task Views, Capacity UI)									
Integration and Internal Testing									
Evaluation, User Feedback, and Refinement									
Final Documentation and Submission Prep									

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